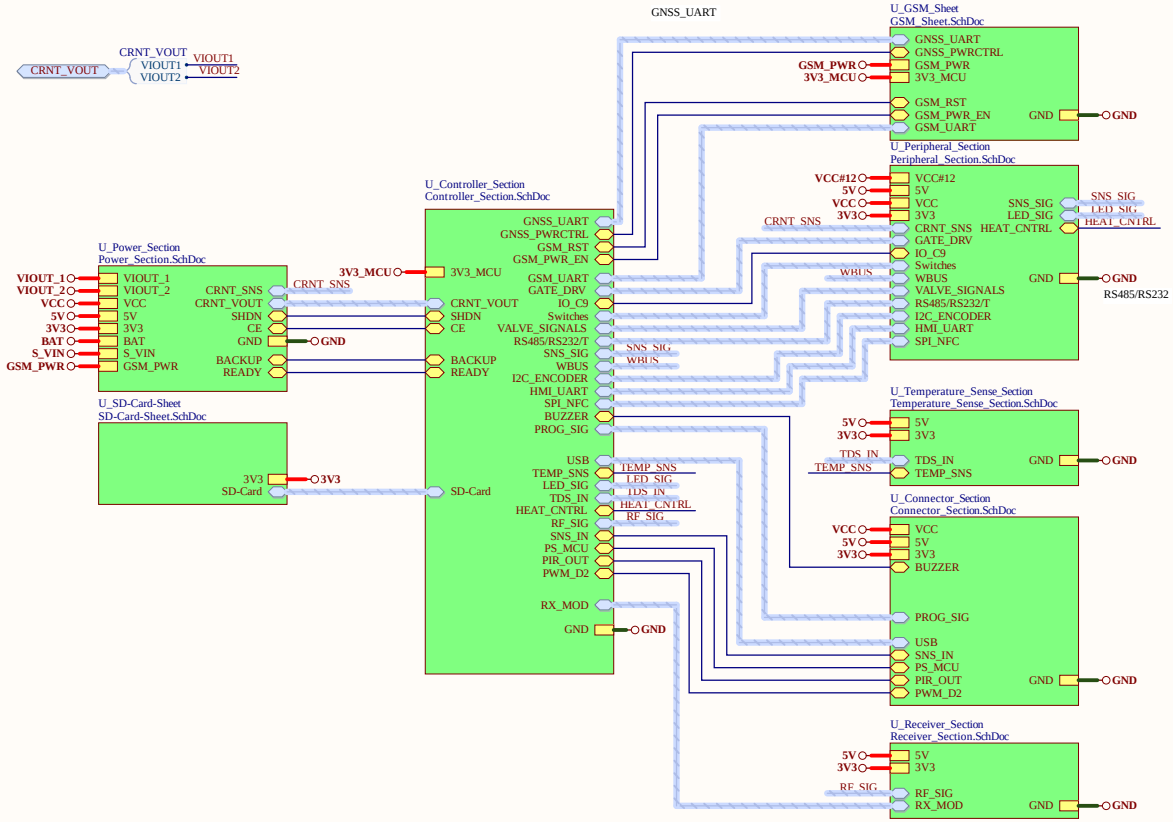
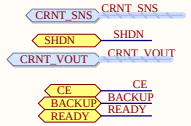
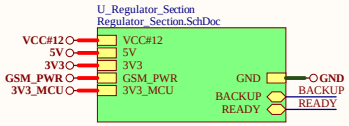
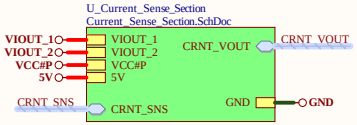


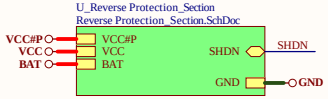
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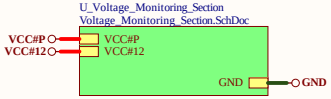
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File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 17

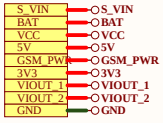


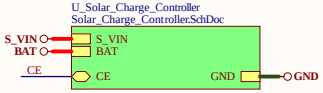










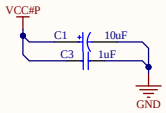


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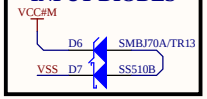
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File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 17

SHDN SHDN

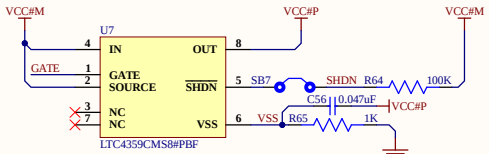
DECOUPLING CAPACITOR



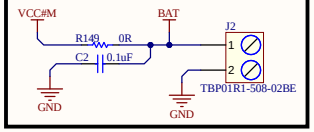
INPUT DIODES



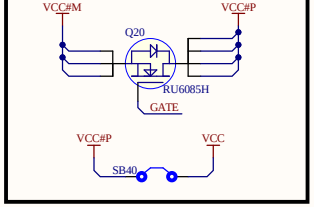
REVERSE POLARITY PROTECTION IC



BATTERY / MAIN INPUT



MOSFET

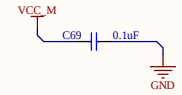


BAT BAT
VCC VCC
VCC#P VCC#P
GND GND

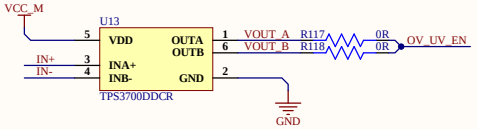
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File does not exist.

Title: Reverse Protection_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 17

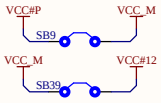
DECOUPLING CAPACITOR



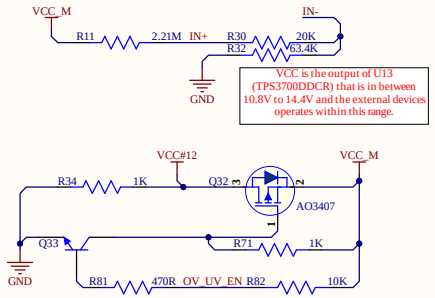
UNDER AND OVER VOLTAGE PROTECTION IC



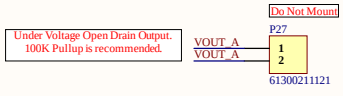
VOLTAGE TO MONITOR



SUPPORTING CIRCUITRY

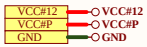


UNDER VOLTAGE OUTPUT



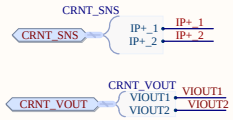
OVER VOLTAGE OUTPUT

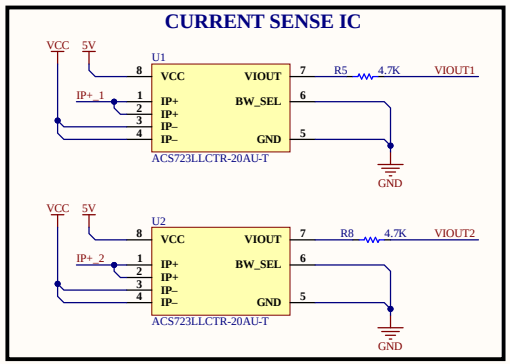




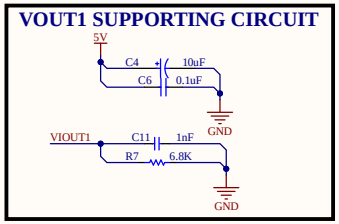
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D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
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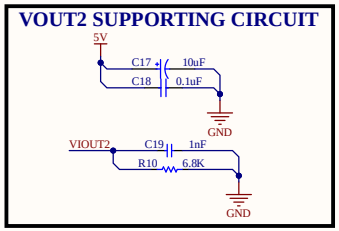
Title: Voltage_Monitoring_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 17

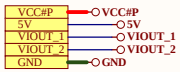




Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.



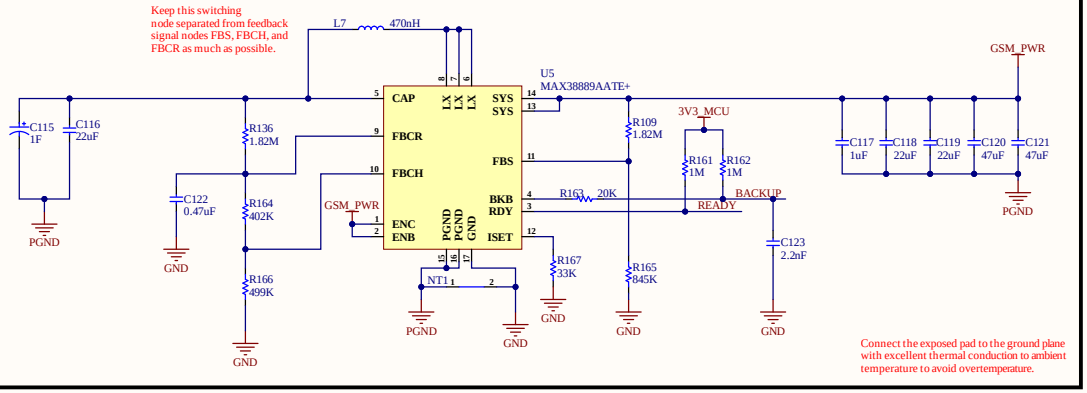




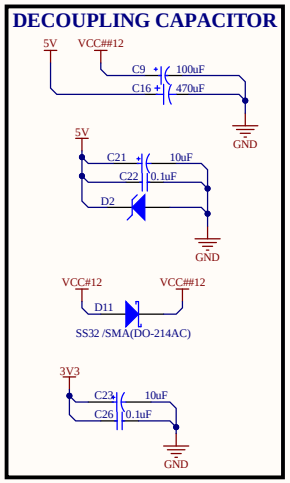
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Title: Current_Sense_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 17

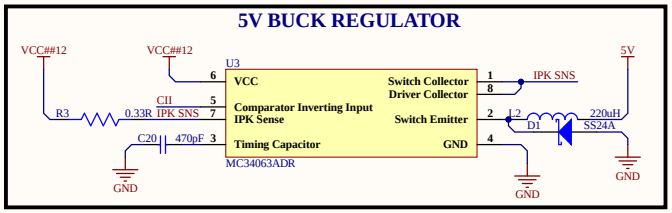
SUPERCAPACITOR BACKUP CONTROLLER



DECOUPLING CAPACITOR

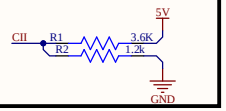


5V BUCK REGULATOR

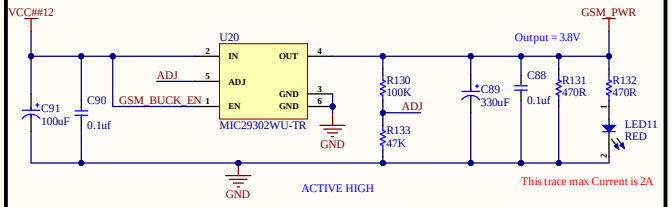


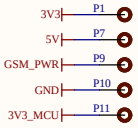
VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

SUPPORTING CIRCUIT

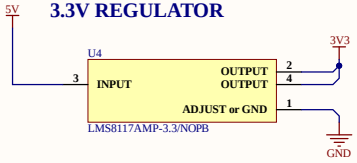


REGULATOR FOR GSM POWER

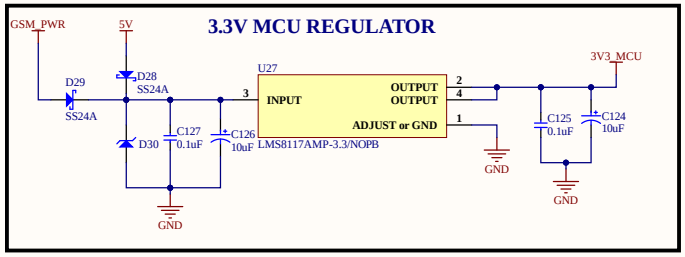


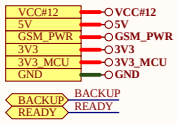


3.3V REGULATOR



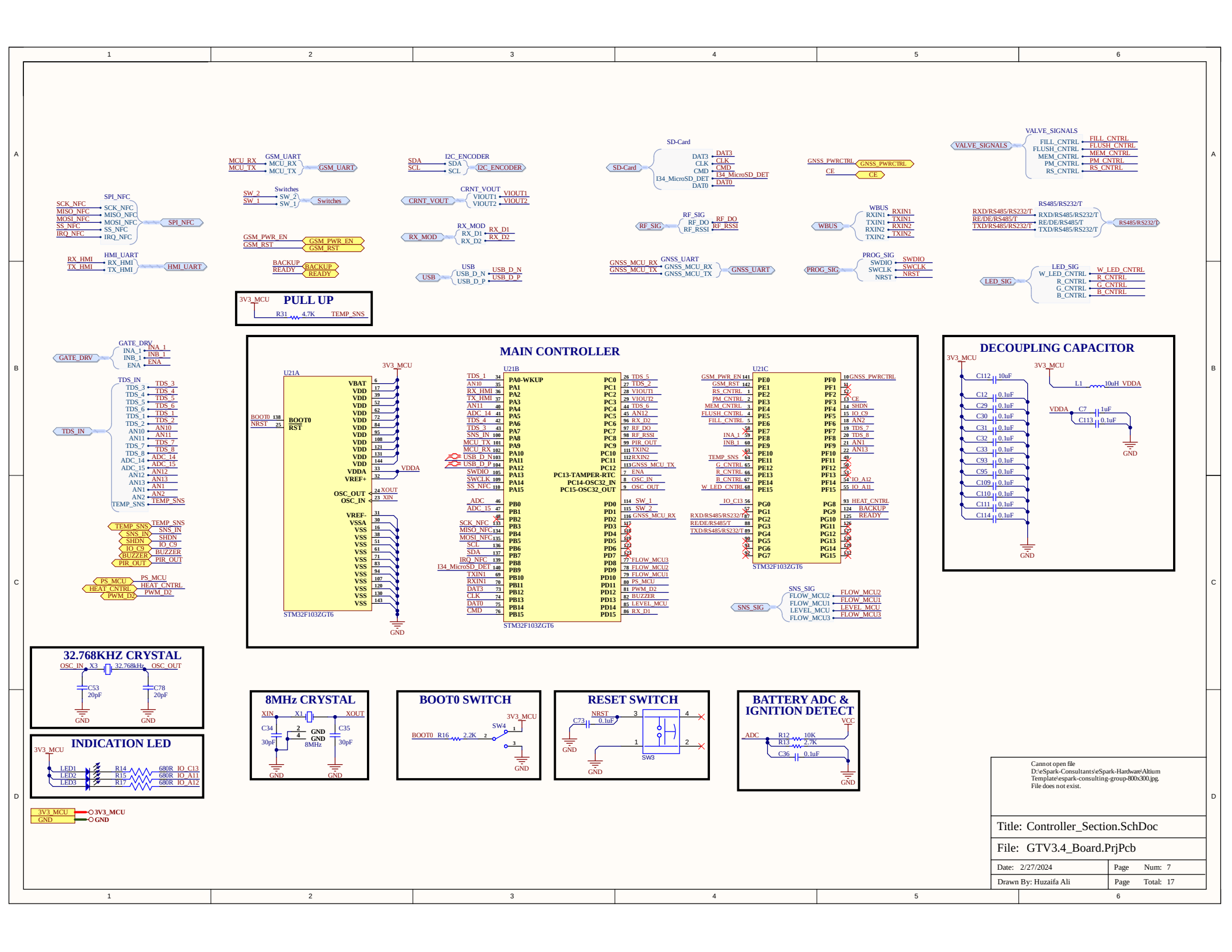
3.3V MCU REGULATOR

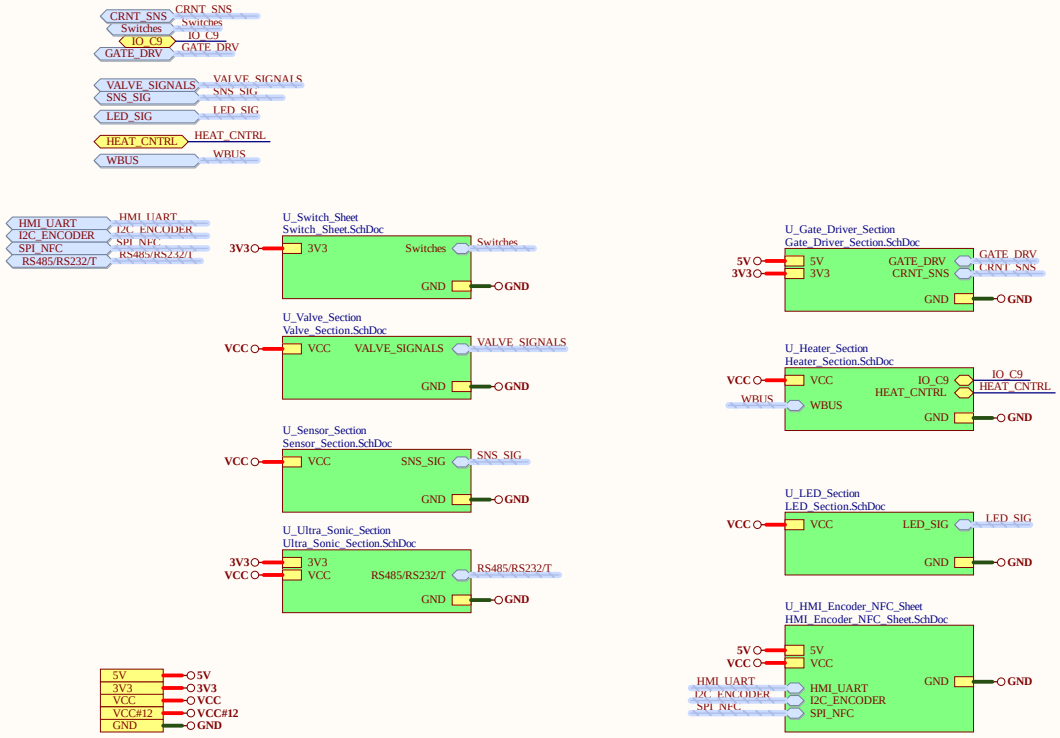




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Title: Regulator_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 2/29/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 17

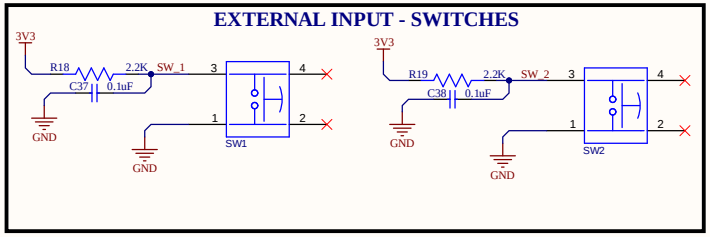

SPI_NFC
SCK_NFC
MISO_NFC
MOSI_NFC
SS_NFC
IRQ_NFC
HMI_UART
RX_HMI
TX_HMI
GSM_UART
MCU_RX
MCU_TX
Switches
SW_2
SW_1
GSM_PWR_EN
GSM_RST
BACKUP
READY
I2C_ENCODER
SDA
SCL
CRNT_VOUT
VIOUT1
VIOUT2
RX_MOD
RX_D1
RX_D2
USB
USB_D_N
USB_D_P
SD-Card
DAT3
CLK
CMD
I34_MicroSD_DET
DAT0
GNSS_PWRCTRL
CE
VALVE_SIGNALS
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL
RF_SIG
RF_DO
RF_RSSI
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
RS485/RS232/T
RXD/RS485/RS232/T
RE/DE/RS485/T
TXD/RS485/RS232/T
PROG_SIG
SWDIO
SWCLK
NRST
GNSS_UART
GNSS_MCU_RX
GNSS_MCU_TX
LED_SIG
W_LED_CNTRL
R_CNTRL
G_CNTRL
B_CNTRL
PULL UP
3V3_MCU
R31
4.7K
TEMP_SNS
GATE_DRV
INA_1
INB_1
ENA
TDS_IN
TDS_3
TDS_4
TDS_5
TDS_6
TDS_1
TDS_2
AN10
AN11
TDS_7
TDS_8
ADC_14
ADC_15
AN12
AN13
AN1
AN2
TEMP_SNS
SNS_IN
SHDN
IO_C9
BUZZER
PIR_OUT
PS_MCU
HEAT_CNTRL
PWM_D2
MAIN CONTROLLER
U21A
U21B
U21C
VBAT
VDD
VSS
VDDA
VREF+
VREF-
VSSA
BOOT0
RST
OSC_OUT
OSC_IN
XOUT
XIN
PA0-WKUP
PC13-TAMPER-RTC
PC14-OSC32_IN
PC15-OSC32_OUT
STM32F103ZGT6
GND
DECOUPLING CAPACITOR
C112
C12
C29
C30
C31
C32
C33
C93
C95
C109
C110
C111
C114
C113
C7
L1
10uH_VDDA
1uF
0.1uF
32.768KHZ CRYSTAL
X3
32.768kHz
C53
C78
20pF
INDICATION LED
LED1
LED2
LED3
R14
R15
R17
680R_IO_C13
680R_IO_A11
680R_IO_A12
8MHz CRYSTAL
X1
C34
C35
8MHz
BOOT0 SWITCH
BOOT0_R16
2.2K
SW4
RESET SWITCH
NRST
C73
SW3
BATTERY ADC & IGNITION DETECT
ADC
VCC
R12
10K
R13
2.7K
C36
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File does not exist.
Title: Controller_Section.SchDoc
File: GTV3.4_Board.PrjPcb
Date: 2/27/2024
Page Num: 7
Drawn By: Huzaifa Ali
Page Total: 17



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File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 17

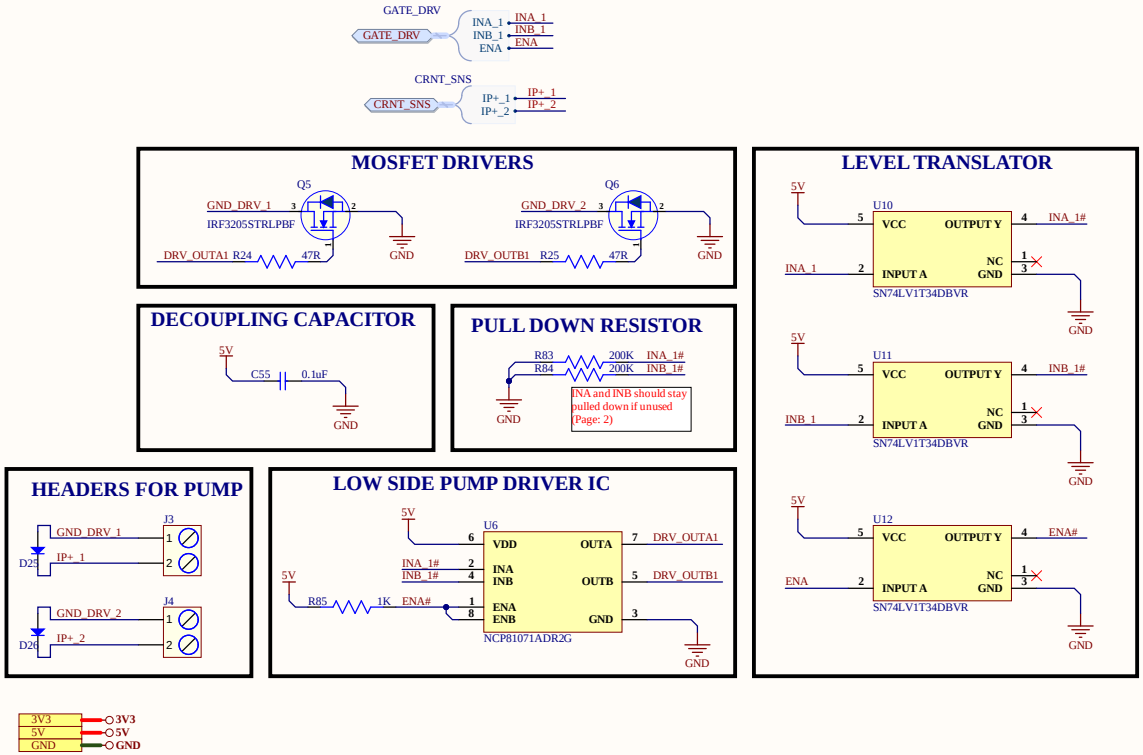






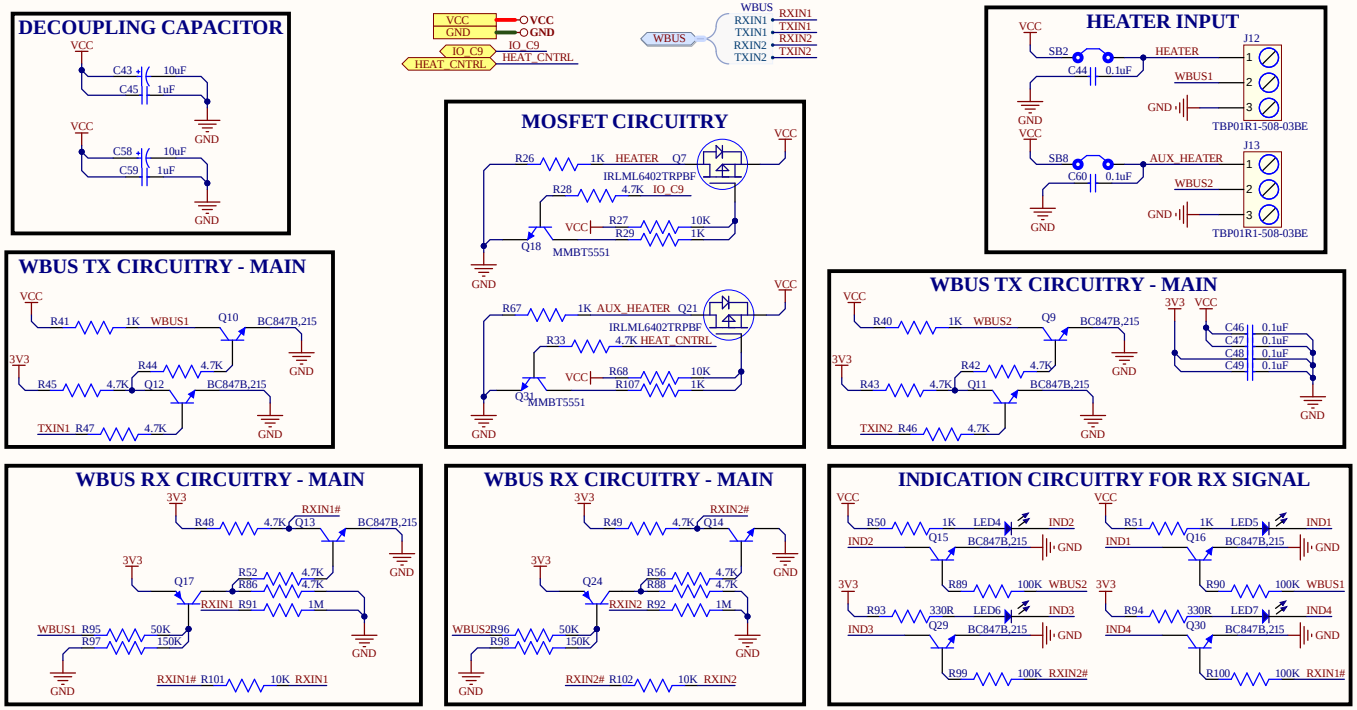
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Title: Switch_Sheet.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 17



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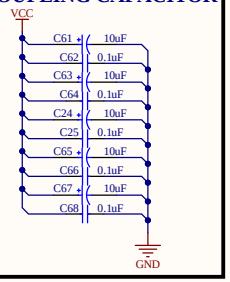
Title: Gate_Driver_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 17



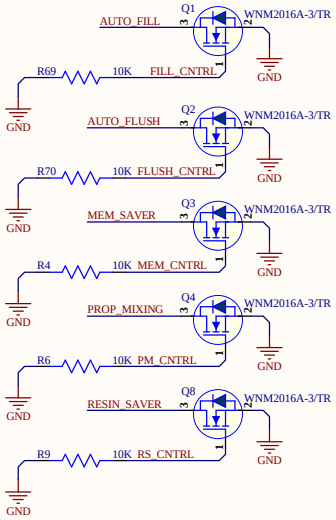
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Title: Heater_Section.SchDoc	
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Date: 2/27/2024	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 17

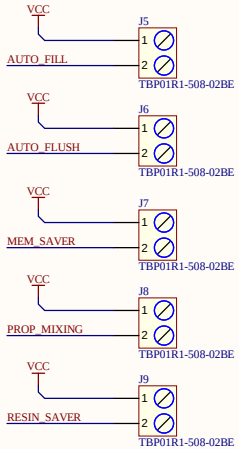
DECOUPLING CAPACITOR



SOLENOID CONTROL CIRCUIT



SOLENOID VALVE INPUTS



VALVE_SIGNALS

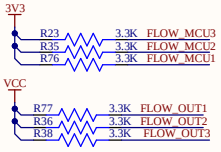
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL

VCC
GND

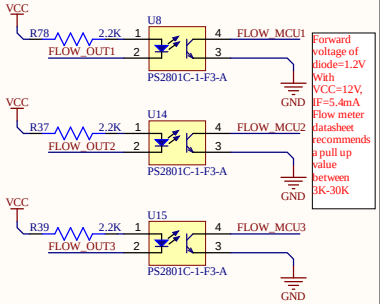
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File: GTV3.4_Board.PrjPcb	
Date: 2/27/2024	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 17

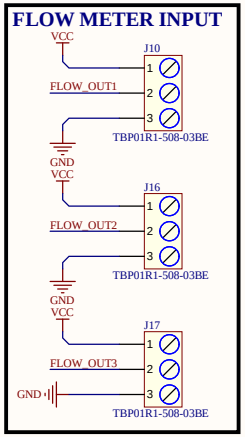
PULL UP RESISTORS



OPTO ISOLATION FOR FLOW METER



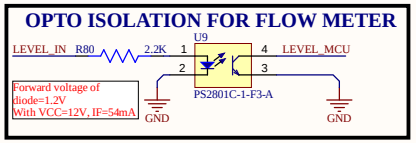
FLOW METER INPUT



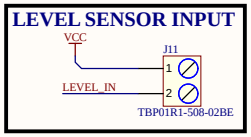
SUPPORTING CIRCUIT



OPTO ISOLATION FOR FLOW METER



LEVEL SENSOR INPUT

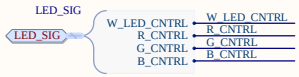


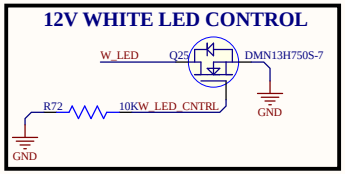


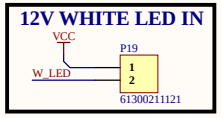


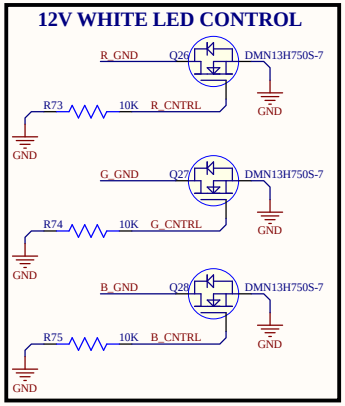
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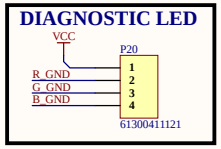
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Date: 2/27/2024	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 17







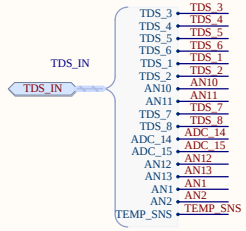




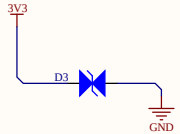


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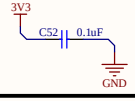
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Date: 2/27/2024	Page Num: 14
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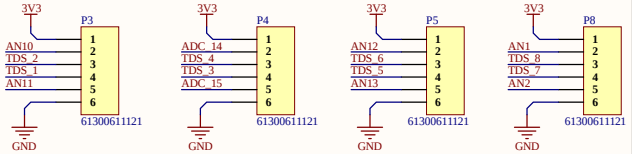
PULL UP RESISTORS



DECOUPLING CAP

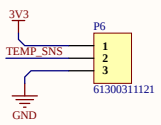


4 X TDS INPUTS



These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

TEMP SENSOR DS18B20

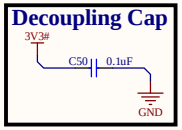




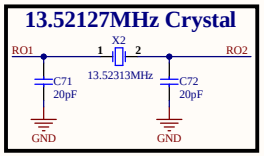
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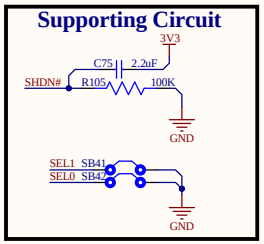
Decoupling Cap

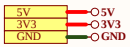


13.52127MHz Crystal



Supporting Circuit

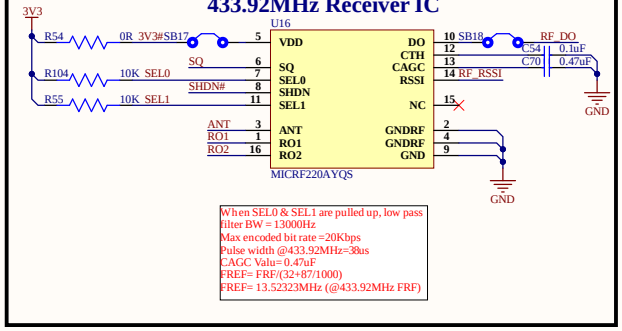




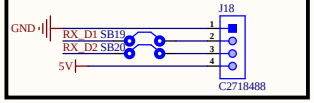




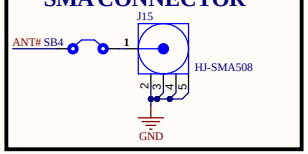
433.92MHz Receiver IC



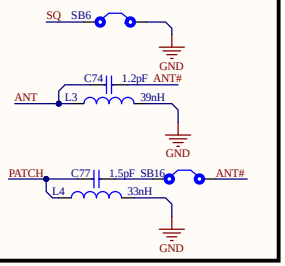
RX 433.93MHz MODULE



SMA CONNECTOR

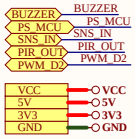


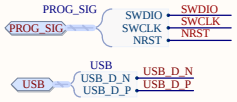
Z-MATCHING CIRCUITRY

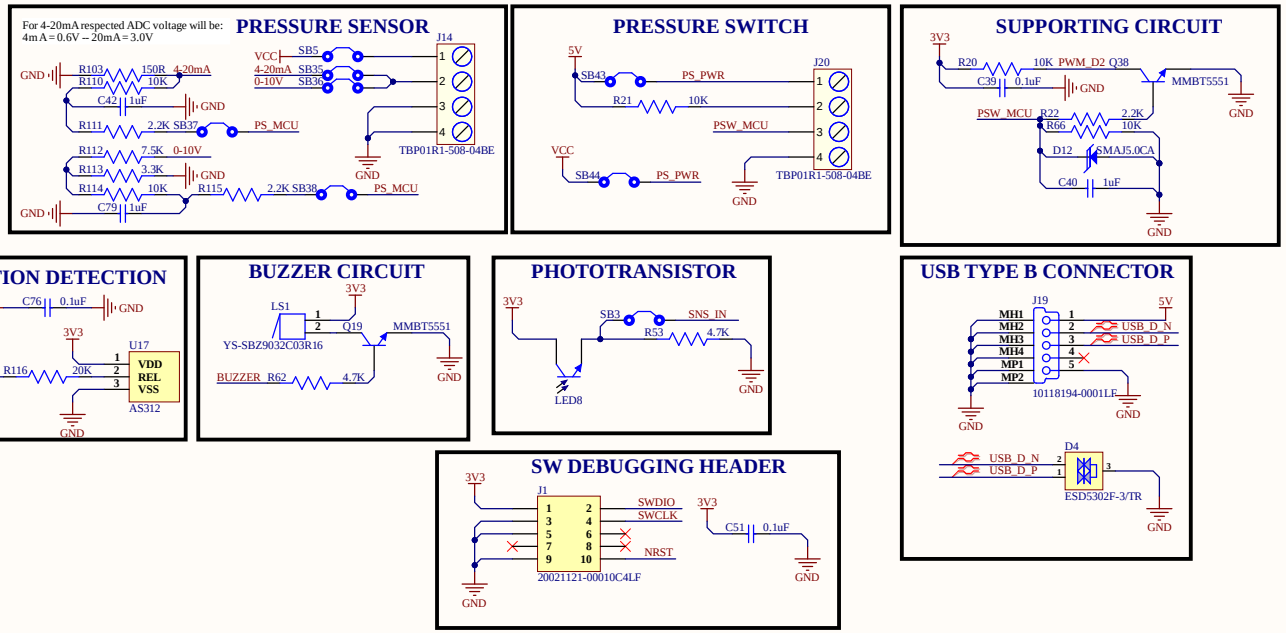


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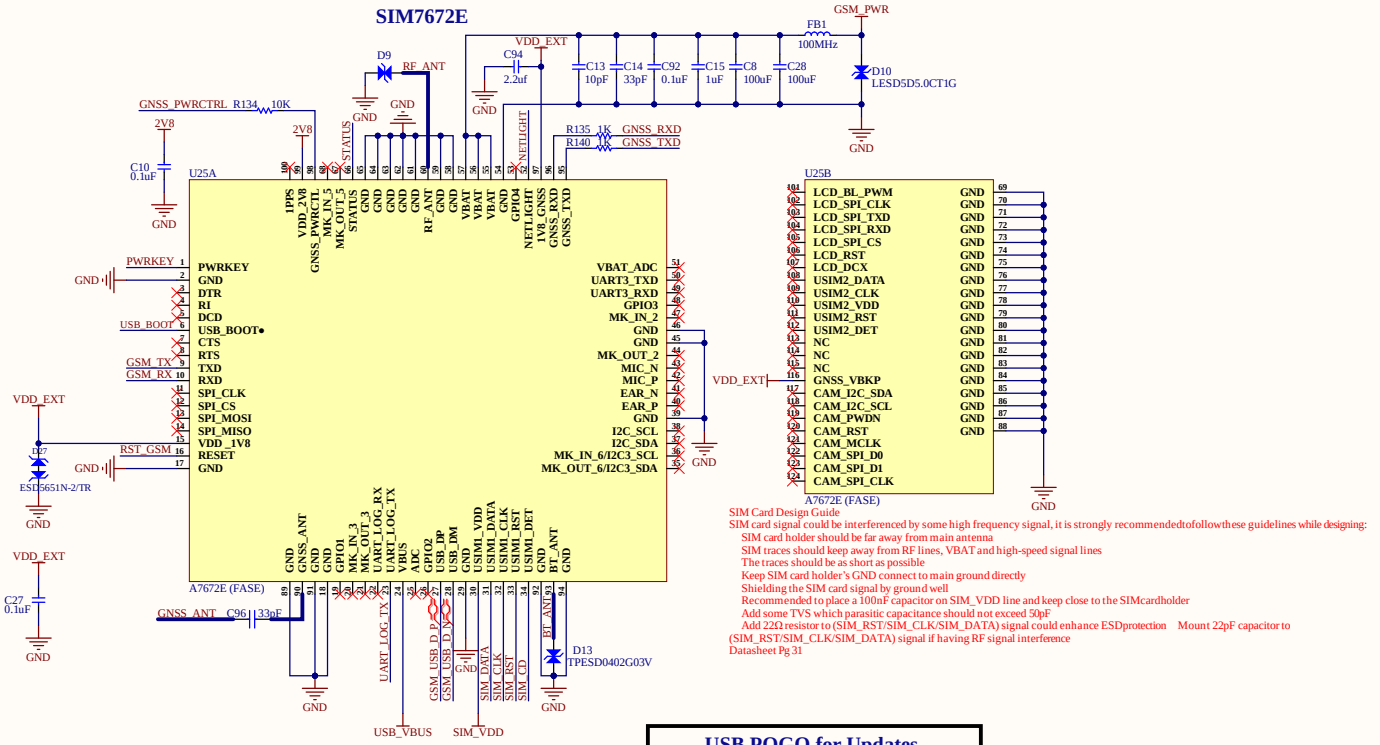




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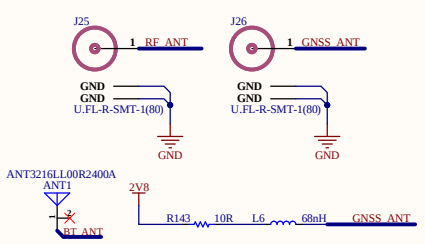
SIM7672E



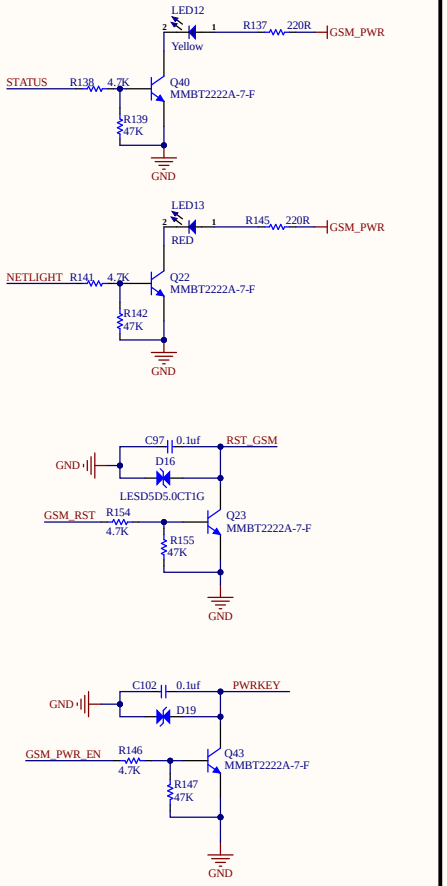
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

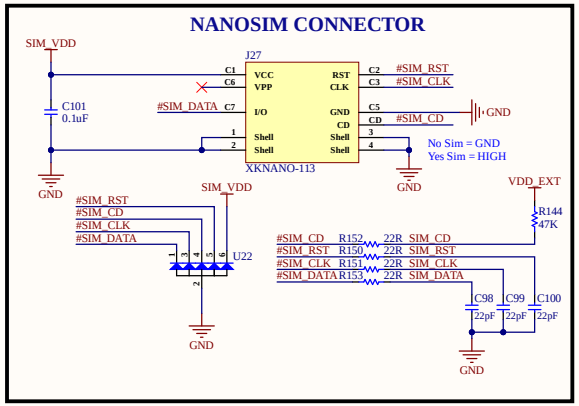
ANTENNA



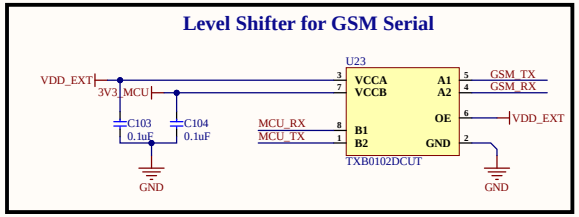
GSM_COMMANDS



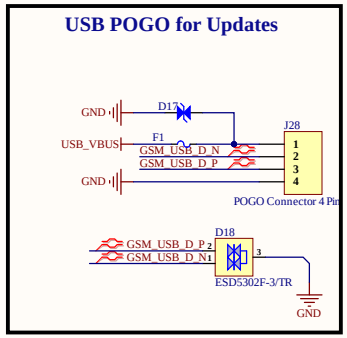
NANOSIM CONNECTOR



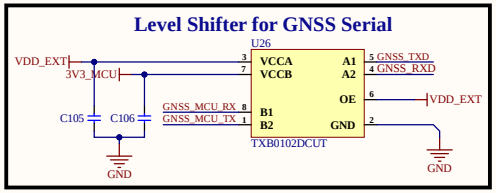
Level Shifter for GSM Serial

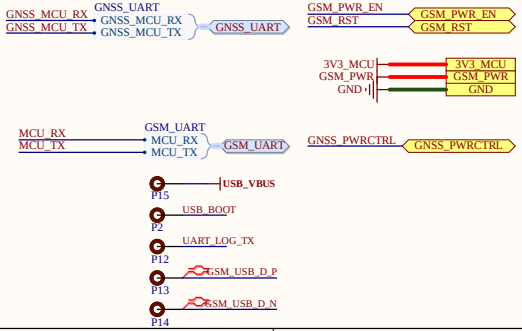


USB POGO for Updates



Level Shifter for GNSS Serial

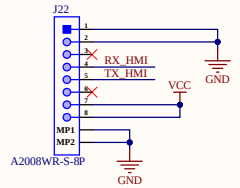




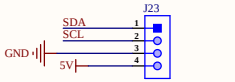
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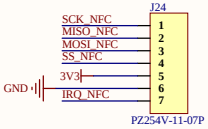
HMI INPUT STONE

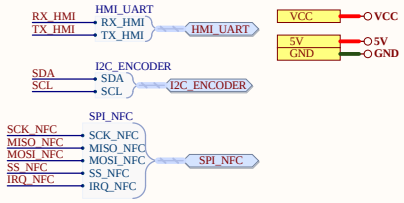


ENCODER INPUT



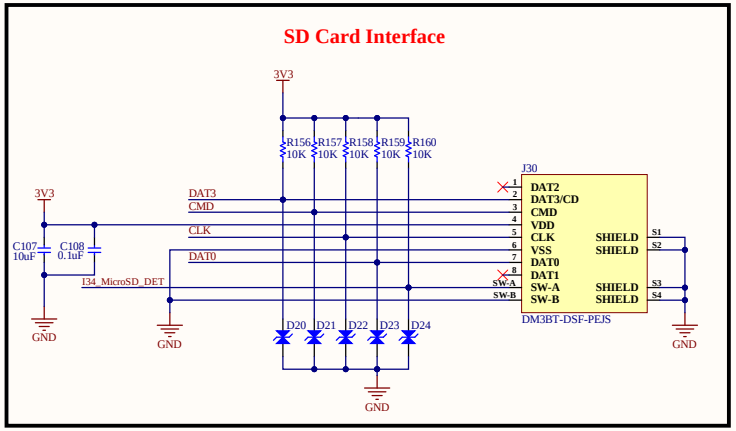
NFC INPUT

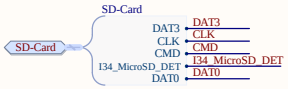




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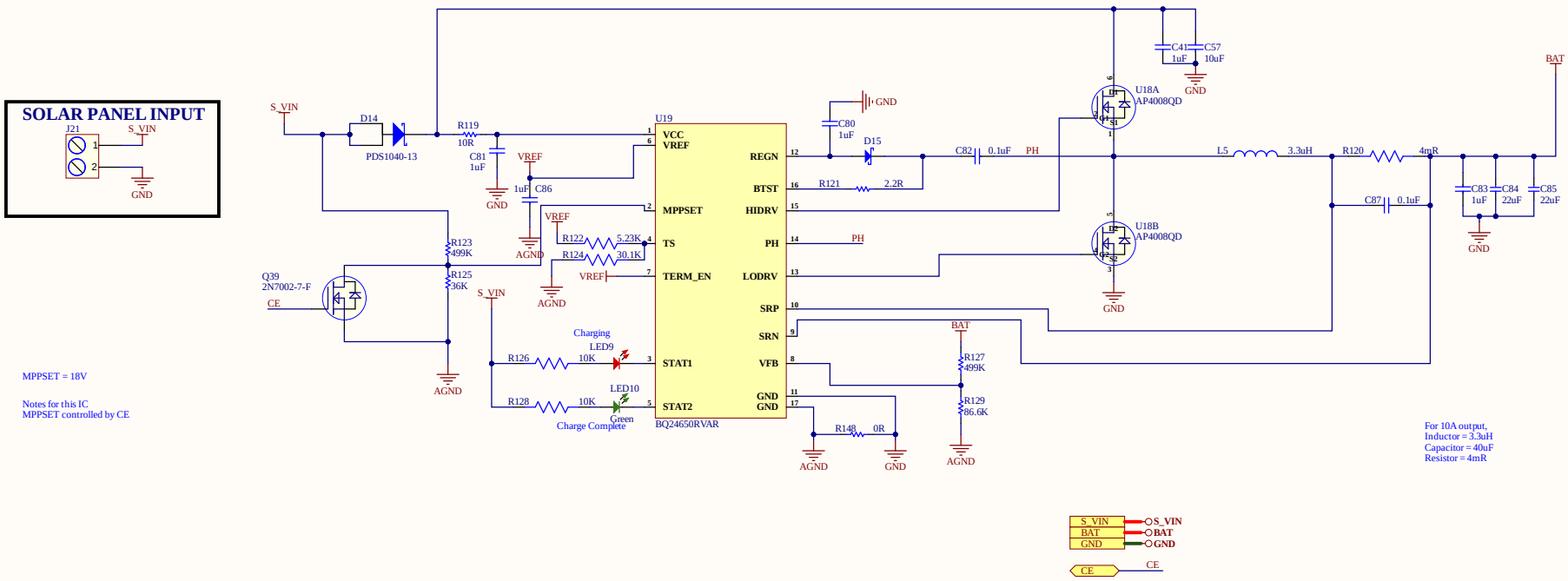






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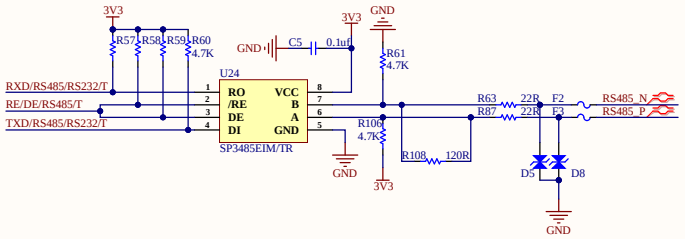
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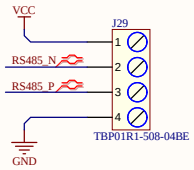
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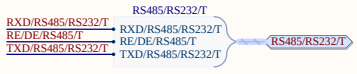
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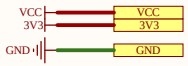
RS485 to UART



ULTRA-SONIC SENSOR







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